

Styrene-Assisted Melt Free Radical Grafting of Glycidyl Methacrylate onto Polypropylene

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ABSTRACT: The free radical grafting reactivity of glycidyl methacrylate (GMA) onto polypropylene (PP) in the molten state is low. This article shows that adding styrene as a second monomer (or comonomer) increases both the rate and yield of GMA grafting and reduces PP chain scission. The proposed mechanism is that when St is added to the PP/GMA/peroxide grafting system, St reacts first with PP macroradicals to form stable styryl macroradicals. These latter then react (or copolymerize) with GMA to form GMA grafted PP. © 1998 John Wiley & Sons, Inc. *J Polym Sci A: Polym Chem* 36: 1053–1063, 1998

Keywords: free radical grafting; polypropylene; glycidyl methacrylate; styrene; maleic anhydride

INTRODUCTION

It has been common practice to use the free radical grafting chemistry to functionalize otherwise apolar and/or chemically inert polymers like polypropylene, polyethylene, and copolymers thereof. Glycidyl methacrylate (GMA) is a very interesting monomer because of its bifunctional character: a free radically reactive double bond together with an epoxy group that can react with a number of other functional groups such as carboxyl, hydroxyl, anhydride, and amine.

However, the melt free radical grafting reactivity of GMA is low. Unless a large excess of a peroxide is employed, the conversion of GMA monomer to grafted GMA onto PP usually does not exceed 10%.^{1–4} Our early work showed that addition of a second monomer and styrene (St) in particular greatly increased the GMA grafting yield together with reduced PP chain scission.^{1,2} The idea of using St as a comonomer originated from a detailed analysis of the mechanism of free radical grafting.

As described earlier,⁵ a free radical grafting process starts with formation of macroradicals along polymer chains by a so-called hydrogen abstraction mechanism. These macroradicals may subsequently follow two competing pathways. They can either initiate the grafting of the monomer (which is desirable) or undergo branching, crosslinking, or chain scission (which is often undesirable). This latter depends very much on the nature of the polymer backbone. In the case of PP, the main side reaction is β -scission associated with PP tertiary macroradicals, which reduces the molecular weight of the polymer. To obtain high grafting yields with reduced chain scission, it is essential that the macroradicals react with the grafting monomer before they undergo side reactions. Because GMA is not reactive towards PP macroradicals, it would be helpful to use a second monomer that can react with them much faster than GMA, and the resulting free radicals can then copolymerize readily with GMA. In this way, instead of grafting GMA directly onto PP chains, the comonomer (St) serves as a mediator to bridge the gap between the PP macroradicals and GMA monomer. With this idea in mind, we came up with styrene as a comonomer, which was found to be very useful for the GMA grafting.^{1,2}

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The main objective of this article is to further substantiate the above stated idea (or mechanism) with well-controlled experiments.

EXPERIMENTAL

Materials

All chemicals (PP, GMA, and St) and experimental procedures in this study were the same as those in ref. 1 or 2, unless specified otherwise. The half-lifetimes of the peroxides are gathered in Table I. For a typical experimental run, a peroxide was added to the liquid monomer(s). The resulting mixture was then premixed with PP porous pellets (45 g) in a small bottle for about 15 min so the liquid reagents could be absorbed. The whole system was charged to a batch mixer of type Haake Rheocord (capacity: 50 cm³), which was preheated up to 180 or 200°C. There were two sigma-shaped rotors in the chamber to ensure mixing. Their rotation speed was 64 turns per minute (rpm), unless stated otherwise. Samples were taken out from the mixing chamber after 15 min of reaction and quenched in liquid nitrogen to stop the reaction. For kinetic studies, additional samples were taken at chosen time intervals for a total duration of about 60 min.

Sample Purification Procedures^{1,2}

There are three types of GMA in the grafting system: grafted GMA onto PP, unreacted or residual GMA, homopolymerized, and/or copolymerized GMA (when styrene is present).

To quantify them, samples were purified in two ways: they were either dissolved in hot xylene and then precipitated in methanol at room temperature; or they were dissolved in hot xylene and then precipitated in acetone at room temperature. The use of methanol allowed to precipitate not only the modified PP but also poly(glycidyl methacrylate), polystyrene and poly(glycidyl methacrylate-co-styrene), which might be formed during grafting. On the other hand, acetone precipitated PP well, and it was capable of dissolving certain amounts of poly(glycidyl methacrylate), polystyrene and poly(glycidyl methacrylate-co-styrene). Those two ways of purifying modified PP samples allowed determination of, with some uncertainties though, the relative importance of homo- and/or copolymerization under the said grafting conditions.

Characterization of Modified PP

The molecular weight of modified PP was measured by size exclusion chromatography at 145°C using 1,2,4-trichlorobenzene as the solvent.

The grafting yields of GMA and/or St onto PP were determined by FTIR. In Figure 1 is shown an IR spectrum of a GMA and St modified PP. It is characterized by three peaks: the one at 2722 cm⁻¹ characteristic of the PP skeleton, the one at 1730 cm⁻¹ corresponding to the ester group of the grafted GMA, and the one at 700 cm⁻¹, which is related to the grafted styrene. The first peak was chosen as an internal reference and the other two peaks were used to determine the grafting yields of GMA and St, respectively. The calibration curves reported in our previous work^{1,2} were used for the quantitative measurements of grafted GMA and St onto PP. The one for grafted GMA was constructed with modified PP samples whose GMA grafting yields were measured by elemental analysis based on the oxygen atoms of GMA. The one for grafted St was built up with the virgin PP mixed with known amounts of polystyrene. In this study, these two calibration curves were further validated with modified PP samples whose GMA and St grafting yields were measured by ¹H-NMR at 80°C using dichloroethane-d₂ as solvent. (see Fig. 2).

RESULTS

In what follows, the contribution of St to the GMA grafting will be presented in terms of GMA grafting yield, PP molecular weight, and grafting kinetics. Also shown will be the effects of the type and concentration of peroxide and that of reaction temperature on the GMA grafting. For the sake of clarity, the grafting systems of PP/GMA/peroxide, PP/St/peroxide, and PP/GMA + St/peroxide will be denoted as GMA system, St system, and GMA + St system, respectively.

Contribution of St to the GMA Grafting Yield Increase

The melt free radical grafting reactivity of GMA onto PP is low. As shown in Figure 3, the GMA grafting yield first increases with increasing initial amount of GMA and then levels off at a weak value. Under the reaction conditions specified in Figure 3, the ultimate GMA grafting yield does not exceed 0.5 phr (parts per hundred parts of

Table I.

Code	Chemical structure	$t_{1/2}(s)$ at 180°C	$t_{1/2}(s)$ at 200°C
DYBP	$t-C_4H_9-OO-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-C\equiv C-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-OO-t-C_4H_9$	85	15.5
DHBP	$t-C_4H_9-OO-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-CH_2CH_2-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-OO-t-C_4H_9$	34	6.0
DIPP	$t-C_4H_9-OO-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-\text{C}_6\text{H}_4-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-OO-t-C_4H_9$	33	6.0
DCUP	$\text{C}_6\text{H}_5-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-OO-\overset{\overset{CH_3}{ }}{\underset{\underset{CH_3}{ }}{C}}-\text{C}_6\text{H}_5$	29	5.2
BP	$\text{C}_6\text{H}_5-\overset{\overset{O}{ }}{C}-O-O-\overset{\overset{O}{ }}{C}-\text{C}_6\text{H}_5$	1.2	0.3

resin; a GMA grafting yield of 0.5 phr corresponds to 0.5 g GMA per 100 g PP), regardless of the initial amount of GMA charged. By contrast, the free radical grafting reactivity of St is much greater under the same conditions as the St grafting yields are much greater. When St is added to the PP/GMA/peroxide system as a comonomer, the GMA grafting yield is dramatically increased. For example, while the GMA grafting yield of the

GMA system is merely 0.45 phr for 6.0 phr GMA charged, it is increased to 1.6 phr in the case of the GMA + St system. Also, the GMA grafting yield no longer levels off, but increases almost linearly with increasing initial amount of GMA. This implies that the initial amount of St charged to the GMA system is an important parameter that controls the GMA grafting yield.

The ability for St to promote the GMA grafting yield is better illustrated in Figure 4, in which is plotted the GMA grafting yield as a function of the molar ratio of $[St]_i/[GMA]_i$ for a given amount of GMA charged (6.0 phr). The GMA grafting yield depends very much on the amount of St added. For example, when 6.0 phr GMA is charged, the amount of the grafted GMA can be as low as 0.45 phr (without St) and as high as 1.6 phr (for a molar ratio of $[St]_i/[GMA]_i = 1.5$ mol/mol).

It should be noted, however, that while adding St promotes the GMA grafting it also causes more polymerization. As a matter of fact, while the amount of the grafted GMA increases almost linearly with increasing $[St]_i/[GMA]_i$, that of the polymerized GMA increases to a greater extent, in an exponential manner. Under the specified reaction conditions, the amount of the polymerized GMA starts to exceed that of the grafted GMA when $[St]_i/[GMA]_i$ is about 1 mol/mol.

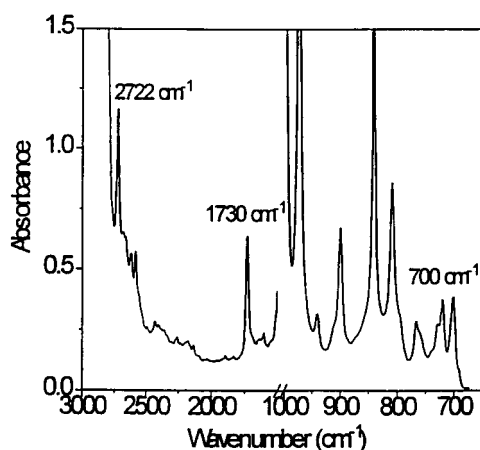


Figure 1. FTIR spectrum of a GMA and St modified PP. Thickness of the film of the modified PP: 100 to 120 μm .

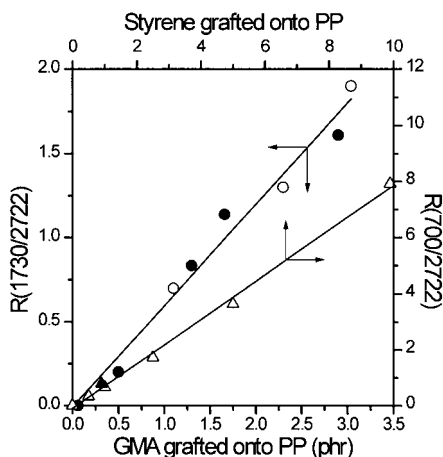


Figure 2. Infrared calibration curves for the quantitative measurements of GMA and St grafted onto PP. (●) Amounts of grafted GMA measured by elemental analysis based on oxygen atom (after ref. 1); (○) Amounts of grafted GMA measured by ^1H -NMR at 80°C using dichloroethane- d_2 as a solvent (this study); (Δ) Amounts of grafted St measured with known amounts of polystyrene in the virgin PP (after ref. 1); ($\blacktriangle$) Amounts of grafted St measured with ^1H -NMR at 80°C using dichloroethane- d_2 as solvent (this study).

Contribution of St to Reduced PP Chain Degradation

A second important contribution of using St as comonomer lies in its ability to reduce β -scission-induced PP chain degradation. In Figure 5 are shown the number- and weight-average molecular weights of GMA-modified PP as a function of

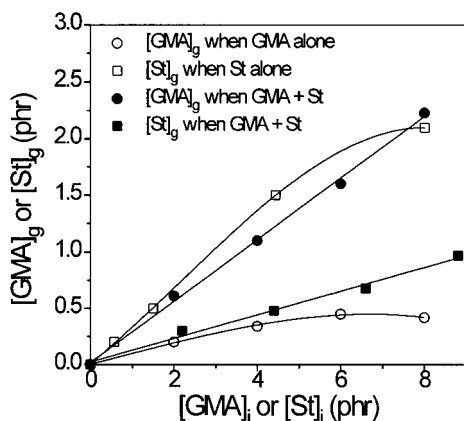


Figure 3. GMA grafting yield as a function of the initial amount of GMA charged with or without St. (○) grafted GMA and (□) grafted St for GMA and St alone, respectively; (●) grafted GMA and (■) grafted St for a mixture of GMA and St, $[\text{St}]_i/[\text{GMA}]_i = 1.5 \text{ mol/mol}$; $[\text{DHBP}]_i = 0.2 \text{ phr}$; temperature = 180°C .

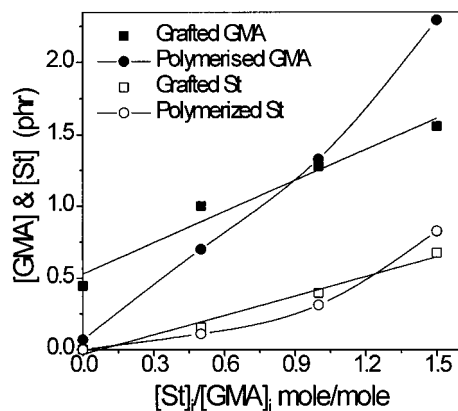


Figure 4. Effect of the molar ratio $[\text{St}]_i/[\text{GMA}]_i$ on the grafting and polymerization of GMA and St. (■) grafted GMA; (●) polymerized GMA; (□) grafted St; (○) polymerized St; $[\text{GMA}]_i = 6.0 \text{ phr}$; $[\text{DHBP}]_i = 0.2 \text{ phr}$; temperature = 180°C .

the molar ratio of $[\text{St}]_i/[\text{GMA}]_i$ for a given initial amount of GMA (6.0 phr) at 180 and 200°C . The more St is added, the higher the molecular weights of the modified PP. It appears that grafting temperature has little effect on PP degradation. This is true at least for temperatures between 180 and 200°C .

The reduced PP chain degradation as a result of adding St is also confirmed by an increase in torque during grafting (torque is a crude measure of the viscosity of the grafting system, thus, the molecular weight of PP chains). Figure 6 shows how the torque evolves during the melt free radical grafting of GMA with different concentrations of St. The higher the amount of St added to the system, the further the torque curve is shifted upward. Note also that without St, there is one peak located at the very beginning of the reaction (between 60 and 120 s). This peak corresponds to PP melting. When St is added, a second peak appears somewhere between 180 and 240 s. It becomes more important with higher St concentration. This second peak likely results from the recombination of PP macroradicals, especially that between styryl-styryl or styryl-GMA radicals, leading to branched or eventually cross-linked PP chains. As such, the molecular weight of the modified PP could be even higher than that of the virgin PP if the branching and/or crosslinking becomes predominant with respect to β -scission. This depends primarily on the concentration of St. As shown in Figure 6, the final torque of the GMA + St system increases with increasing $[\text{St}]_i/[\text{GMA}]_i$. When this latter is 1.5 mol/mol, the

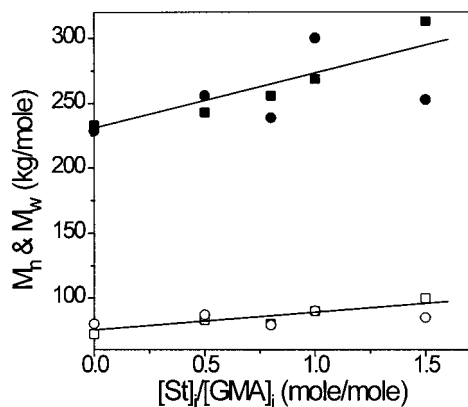


Figure 5. Number- and weight-average molecular weights, $\bar{M}_n$ and $\bar{M}_w$ of GMA modified PP as a function of $[\text{St}]_i/[\text{GMA}]_i$. (●,■): $\bar{M}_w$; (○,□): $\bar{M}_n$; (○,●): 180°C; (□,■): 200°C. The data at 200°C are taken from ref. 1. $[\text{GMA}]_i = 6.0$ phr, $[\text{DHBP}]_i = 0.2$ phr.

final torque of the grafting system is more than twice as high as that of the GMA system and is close to that of the virgin PP. In this study, the GMA + St-modified PP samples in the presence of St are not crosslinked, as they are all soluble in hot xylene. In other words, no gel was found in those GMA + St-modified PP samples. These results suggest that free radically grafting St onto PP could be an interesting way to create long chain branching PP.

The free radical grafting of GMA onto PP is expected to alter the polarity of the original polymer. As such, a GMA-modified PP may become immiscible with the original PP when the grafting yield reaches a certain threshold. One might then attribute the torque increase to a possible phase separation between PP-*g*-GMA and PP. This explanation is, however, not in line with the following facts and/or arguments. First of all, the free radical grafting is quite uniform over the entire system in the batch mixer. GPC coupled with IR showed that all PP chains are almost evenly grafted. In other words, the free radical grafting does not lead to a situation where only a fraction of PP is grafted and the other not. Secondly, when PP is mixed with PP-*g*-GMA in the batch mixer, only one peak is observed, which is located between 60 and 180 s, corresponding to the melting of PP and PP-*g*-GMA.

Contribution of St to the GMA Grafting Rate Increase

In addition to increasing the GMA grafting yield and reducing PP chain degradation, adding St

also speeds up the GMA grafting rate. In Figure 7 are shown the grafting kinetics of the GMA, St, and GMA + St systems. The grafting rate of the GMA system is relatively slow as the amount of the grafted GMA increases gradually with time. That of the St system is, on the other hand, much faster and the ultimate grafting yield is reached within less than 2 min. Note that these 2 min include not only the time of the free radical grafting but also the time necessary for melting the PP and homogenizing the reacting system. In the case of the GMA + St system, both the GMA and St grafting rates become as fast as that of the St system. It is also interesting to note that for the GMA grafting system, when 3.0 phr GMA charged, the GMA grafting yield is about 0.45 phr after 45 min of reaction. For the same weight amount, the St grafting yield of the St system is much higher (0.75 phr). In the case of the GMA + St system ($[\text{GMA}]_i = 3.0$ phr, $[\text{St}]_i/[\text{GMA}]_i = 1.5$ mol/mol), the GMA grafting yield is increased to 0.78 phr, whereas that of St is reduced to 0.3 phr.

Effects of the Nature and Concentration of Peroxide

In Figure 8 is shown the GMA grafting yield as a function of the peroxide concentration for various peroxides whose chemical structures and/or half-lifetimes are different (see Table I). As expected, the higher the peroxide concentration, the higher the GMA grafting yield, regardless of the nature of the peroxide employed. It is noteworthy that among the various peroxides tested, the grafting efficiency of DIPP is the greatest, that of BP is by

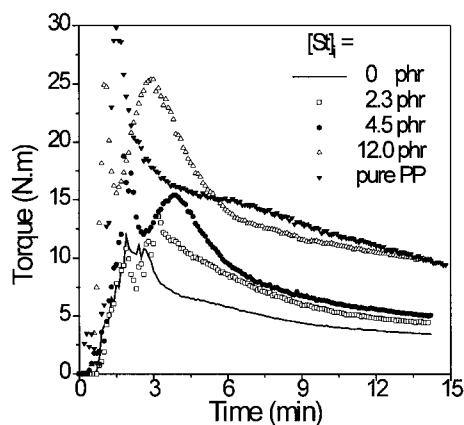


Figure 6. Effect of adding St on the torque as a function of grafting time. $[\text{GMA}]_i = 6.0$ phr, $[\text{St}]_i = 0, 2.3, 4.5$, or 12.0 phr, $[\text{DHBP}]_i = 0.2$ phr, temperature = 180°C.

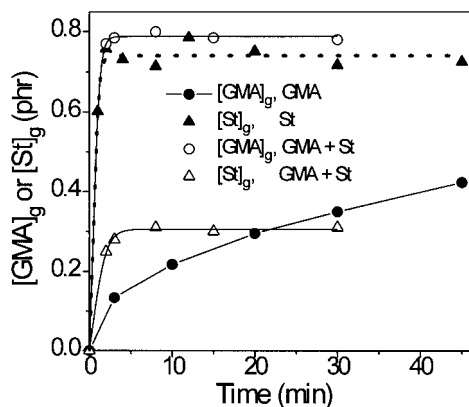


Figure 7. Effect of St on the GMA free radical grafting rate. $[GMA]_i = 3.0$ phr, $[DHBP]_i = 0.3$ phr; 180°C . (●): grafted GMA for the GMA system, (▲): grafted St for the St system, $[St]_i = 3.0$ phr, (○) grafted GMA for the GMA + St system, $[St]_i/[GMA]_i = 1.5$ mol/mol, (△): grafted St for the GMA + St system.

far the lowest and that of DHBP, DYBP, or DCUP lies in between. A poor free radical grafting efficiency of BP is not surprising because of its short half-lifetime (1.2 s at 180°C and 0.3 s at 200°C). A bit higher grafting efficiency of DIPP might be associated with the benzene ring it bears, which could improve its solubility in the molten PP. The difference is insignificant among DHBP, DYBP, and DCUP unless their concentrations are very low, say, below 0.1 phr. These results suggest that there is no obvious correlation between the grafting efficiency and the half-lifetime of a peroxide unless this latter is too short. After all, this is not surprising because the grafting efficiency of a peroxide is dictated not only by its half-lifetime but also by its solubility and its hydrogen abstracting capability.⁵

In Figure 9 is shown the effect of the concentration of a given free radical generator (DHBP) on the melt free radical grafting of the GMA + St system. As expected, the amounts of grafted GMA and St increase with increasing amount of the peroxide added. On the other hand, those of the polymerized GMA and St first increase with an increasing amount of the peroxide, reach a maximum, and then decrease. As for the sum of the grafted and polymerized GMA, it first increases with increasing peroxide concentration and then levels off. The same behavior is observed for St. These results suggest that for a practical application, one should take into account not only the grafting yield and PP chain degradation, but also the amounts of polymerized and nonreacted

monomer. As a rule of thumb, when the amount of the grafting monomer is below 10 phr, it would be good if the amount of the peroxide ranges from 0.2 to 0.4 phr.

Effect of Temperature

The results of Figures 8 and 9 indicate that the GMA grafting does not depend much on temperature, at least as far as the GMA and St grafting yields are concerned. This is further confirmed by the fact that the amounts of grafted GMA at 180 and 200°C are virtually the same for a relatively large peroxide concentration range (see Fig. 10).

DISCUSSION

Mechanism of St-Assisted Free Radical Grafting

The data obtained in this study further substantiate the efficiency of using a comonomer (St) to promote the free radical grafting of GMA onto PP in favor of increased grafting yield (see Figs. 3 and 4), reduced chain scission (see Figs. 5 and 6), and enhanced grafting rate (see Fig. 7). However, one may explain the significant effect of adding St based on the so-called "solvating effect." In fact, one may argue that the solubility of GMA in the molten PP is small because of its polar character. Thus, addition of St could raise its solubility and, consequently, its grafting yield. To check this solvating effect, we replaced St by xylene and ethyl benzene. These two chemicals are expected to have a similar or stronger solvent power than St. No difference was observed between the GMA

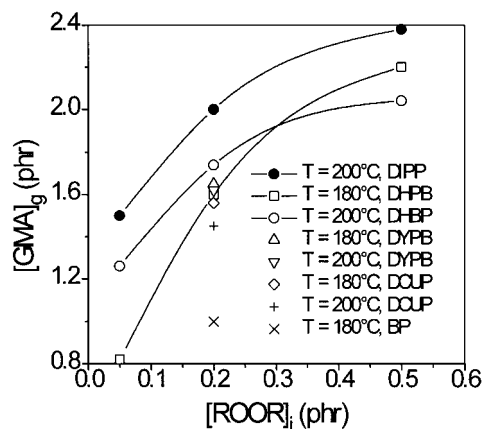


Figure 8. Comparison of the grafting efficiency of various peroxides at 180 and 200°C . $[GMA]_i = 6.0$ phr, $[St]_i/[GMA]_i = 1.5$ mol/mol.

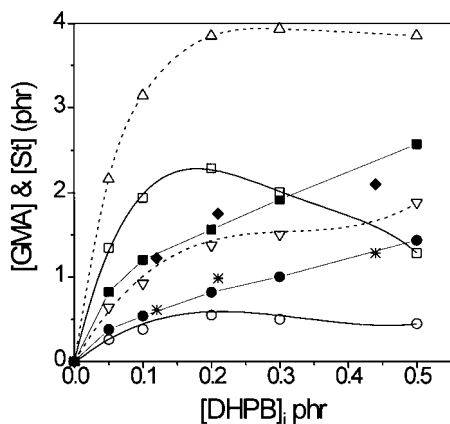


Figure 9. Effects of the peroxide concentration on the melt free radical grafting of GMA + St onto PP. (■) grafted GMA; (●) grafted St; (□) polymerized GMA; (○): polymerized St; (△) total amount of the grafted and polymerized GMA; (▽) total amount of the grafted and polymerized St. $[GMA]_i = 6.0$ phr; $[St]_i = 6.6$ phr; $[St]_i/[GMA]_i = 1.5$ mol/mol, $T = 180^\circ\text{C}$. (◆) grafted GMA and (*) grafted St at 200°C (the data at this temperature are taken from ref. 1).

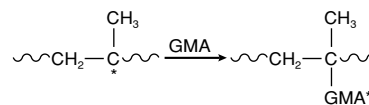
grafting yield of the GMA system and that in the presence of one of these two chemicals.

We also replaced St by α -methyl styrene. It was found that unlike St, α -methyl styrene (α -MeSt) was incapable of improving the GMA grafting yield. This is because α -MeSt is less reactive toward PP macroradicals, due possibly to an accrued steric hindrance. Also, it would be less capable of copolymerizing with GMA. This is confirmed by the fact that no homopolymerized or copolymerized α -MeSt was found in the grafting system. The above discussion leads to the conclusion that the fundamental basis of the comonomer concept is the reactivity ratio of free radical copolymerization. This means that for a monomer to be grafted onto a macromolecular chain, adding a comonomer will be beneficial for improving the monomer grafting yield only if this comonomer reacts with a macroradical more rapidly than the grafting monomer and the resulting macroradical is capable of copolymerizing with the grafting monomer. Otherwise, the presence of this comonomer will likely not have any beneficial effect. Instead, more often than not, adding such a comonomer will be detrimental to the monomer grafting yield because it competes against the grafting monomer for the same macroradicals. This is the scenario of the free radical grafting of ricinloxazoline maleinate (OXA) onto PP.⁶ This monomer cannot copolymerize with St easily. Thus, it cannot be

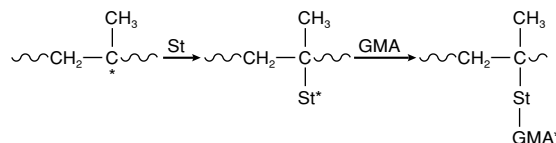
grafted onto PP via styryl macroradicals. Consequently, the OXA grafting yield is not increased but decreased upon addition of St.

To sum up, the mechanisms of the melt free radical grafting of GMA onto PP with and without St may be depicted as follows:

GMA alone:



GMA + St:



When St is present, a small portion of GMA molecules may also be grafted directly onto PP.

Kinetics of St-Assisted Free Radical Grafting

It has been generally believed that the melt free radical grafting rate is dictated primarily by the rate of formation of free radicals. This latter can be characterized by the half-lifetime of the free radical generator involved. The validity of this

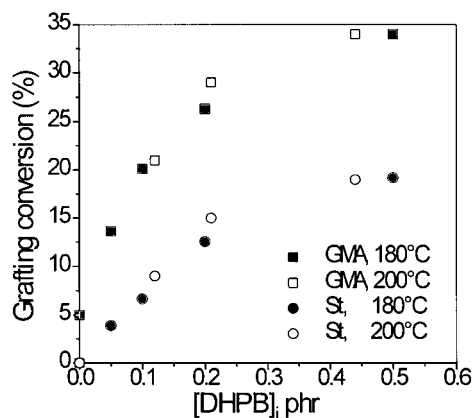


Figure 10. Comparison of the GMA and St conversions (which is defined as the percentage of the amount of GMA monomer converted to grafted GMA) as a function of the peroxide concentration at 180 and 200°C . (■): GMA at 180°C ; (□): GMA at 200°C ; (●): St at 180°C ; (○): St at 200°C . $[GMA]_i = 6.0$ phr, $[St]_i/[GMA]_i = 1.5$ mol/mol.

common belief is questioned for the first time in this study. In fact, we believe that the reaction between the macroradicals and the monomer can also be rate limiting. This is because for a given peroxide (DHBP), the GMA grafting rate in the presence of St is much faster than that without (see Fig. 7). It would be difficult to imagine that the rate of decomposition of DHBP becomes much faster upon adding St! A much faster GMA free radical grafting rate in the presence of St can only be explained by the above-stated reaction mechanism. When St is added, St reacts first with PP macroradicals and the resulting free radicals then copolymerize with GMA. By this mechanism, the GMA grafting yield is increased, the PP chain degradation by β -scission reduced, and the grafting rate accelerated. A higher reactivity of St toward PP macroradicals than GMA might be related to the resonance effect between the double bond and the benzene ring of St. Detailed studies on this subject are needed.

To further substantiate the above analysis, we wanted to see how mixing intensity could affect the free radical grafting rate of the GMA system. The idea was that if the solubility of GMA was indeed small in the molten PP and was the cause of the slow GMA grafting kinetics (Fig. 7), as one might suggest, then the higher the rotation speed of the rotators, the better GMA would be dispersed in the molten PP and the faster it would be grafted onto PP. In Figure 11 is shown the GMA grafting yield as a function of time for three different rotation speeds. It is seen that the rate with which GMA is grafted onto PP is virtually the same, irrespective of mixing intensity that is imposed on the reacting system. This finding further supports the aforementioned conclusion that the solubility argument cannot explain the contribution of adding St on the GMA grafting rate and yield increase.

Nevertheless, several perplexing points remain in this study. The first one is the mismatch between the half-lifetime of DHBP (~ 6 s at 200°C) and the reaction time scale (e.g., 45 min) in which the GMA grafting yield without St keeps increasing (see Fig. 11). One may argue that the value of the half lifetime of DHBP at 200°C given above is calculated by extrapolating those measured at much lower temperatures and under conditions that are quite different from those of the melt free radical grafting.⁵ The real half-lifetime of DHBP could be longer because of the highly viscous medium. In fact, it was found that the rate of decomposition of a peroxide in a polymer melt could be

the same as that in an inert solvent.⁹ In any event, as large as this difference may be, it would not be enough to match the GMA grafting time scale.

Another argument that might be used to explain the above-mentioned mismatch is based on thermally induced GMA free radical grafting. In fact, the grafting system in the batch mixer is constantly exposed to air. At high temperatures, free radicals can be generated without external free radical generator. The supply of the free radicals may be sufficient for the GMA grafting to continue even after the externally furnished free radicals have used up. Separate experiments showed that without externally added free radical generator, the amount of the grafted GMA onto PP was virtually zero. This is related to the fact that the PP used in this study was a commercial product containing additives that were able to neutralize free radicals. The effect of the presence of commercial additives on melt free radical grafting is discussed in detail elsewhere.^{5,7} The second perplexing point is related to the mismatch between the time in which PP gets melted and in which the GMA grafting (in the presence of St) has almost gone to completion and the time in which a second peak appears in the torque curve. Recall that the PP has molten in the first 60 to 120 s, while the second peak of the torque is located between 180 and 240 s (see Fig. 6). On the other hand, the GMA grafting yield reaches its ultimate value within less than 120 s (see Fig. 10). This implies that the intrinsic free radical grafting rate is so much accelerated by the presence of St that the overall grafting kinetics becomes controlled primarily by PP melting. A thorough discussion

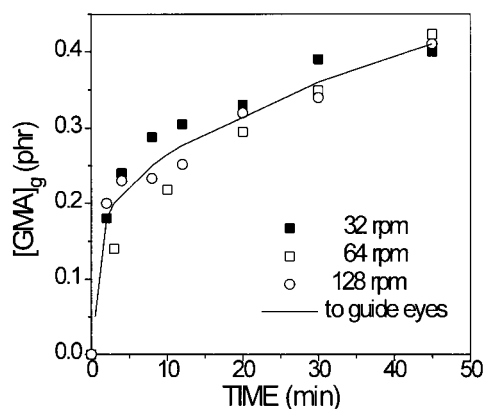


Figure 11. Effect of mixing intensity (rotation speed of the rotors) on the kinetics of grafting GMA alone onto PP. $[\text{GMA}]_i = 3.0$ phr, $[\text{DCUP}]_i = 0.3$ phr, temperature = 200°C .

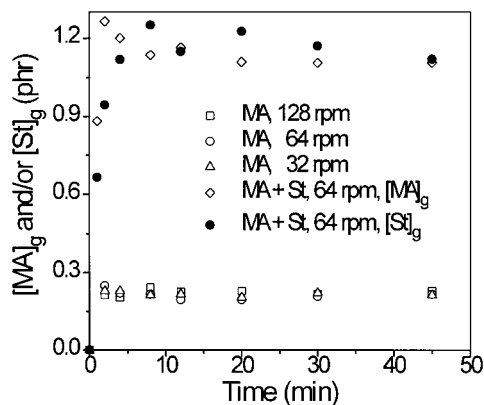


Figure 12. Kinetics of the melt free radical grafting of maleic anhydride (MA) onto PP with or without St. $[MA]_i = 3.0$ phr, $[St]_i/[MA]_i = 0$ or 1.0 mol/mol, $[DCUP]_i = 0.3$ phr, temperature = 200°C.

on this subject is given in a separate article.⁸ The question now is why the second peak of the torque curve appears after the GMA grafting is over. If this peak does indeed result from the termination by recombination of PP macroradicals, it would mean that they are quite stable at high temperatures. This may be particularly so for styryl radicals, considering the resonance effect of the benzene ring. Then why are these long-life styryl-macroradicals no longer able to initiate further GMA grafting (the amount of nonreacted GMA after grafting can be as much as 20 to 40%)? The only scenario would be that the concentration of the macroradicals is so low after 2 min of reaction that they can no longer initiate the GMA grafting. Nevertheless, it is enough to increase the torque after recombination. This is because recombination of a few macroradicals would be enough to see some torque increase.

A third puzzling point happened to appear when we were curious about how fast maleic anhydride (MA) would be grafted onto PP. MA has probably been the most frequently used monomer for functionalizing polymers. However, the kinetic aspect of the problem has rarely been studied. Figure 12 shows the evolution of the MA grafting yield as a function of time with or without St. Interestingly, although the MA grafting yield is increased by a factor of about 4 or 5 in the presence of St, the MA grafting rate without St is as fast as that with St. In both cases, the grafting has gone to completion within 2 min. Moreover, mixing does not have any influence on the MA grafting rate and yield. This is in sharp contrast to what we have seen above for GMA, despite the

fact that without St, the free radical grafting reactivity of both MA and GMA is low (see Figs. 11 and 12) and that the solubility of MA in the molten PP is small.¹⁰ The question remains open as to why the melt free radical grafting of MA is much faster than that of GMA. Work on this subject is underway.

Quantitative Contribution of St to the GMA Grafting Yield

In addition to the three perplexing points discussed above, it is also surprising that when an equal amount of GMA or St is to be grafted separately onto PP, the amount of the grafted St is much higher than that of the grafted GMA. When they are mixed together, however, the amount of grafted St is less than that of grafted GMA (see Fig. 3). This tends to suggest that in the case of the GMA + St system, some of the GMA molecules are grafted directly onto PP, while most of them are through styryl macroradicals. If so, it would be interesting to see whether the increase of the amount of the grafted GMA in the presence of St could be correlated to that of the grafted St.

According to the comonomer concept, a styryl macroradical, once formed, will copolymerize with one or more GMA monomers. For the sake of clarity, we define the increase in the amount of grafted GMA (or the GMA grafting yield gain) as a result of adding St, $[GMA]_g^+$, as the difference between the amount of grafted GMA obtained in the presence of St, $[GMA]_g^{St}$, and that obtained with the GMA system in the absence of St, $[GMA]_g^0$. If we assume that each styryl macroradical copolymerizes with one GMA molecule only and that the amount of the GMA grafted directly onto PP is about the same as $[GMA]_g^0$, then

$$[GMA]_g^+ = [GMA]_g^{St} - [GMA]_g^0 = [St]_g \quad (1)$$

To check the validity of the above relationship, experimental values of $[GMA]_g^+$ are plotted against those of $[St]_g$ and then compared with the prediction. These data fall on the theoretical line surprisingly well (Fig. 13). This implies that the assumptions made above, though crude, may not be far from the real picture of the grafting mechanism and the architecture of the grafts. A few grafts would contain one single GMA moiety, while a great majority of them would be composed of one GMA and one St.

However, it should be pointed out that despite a good agreement between the theoretical predictions

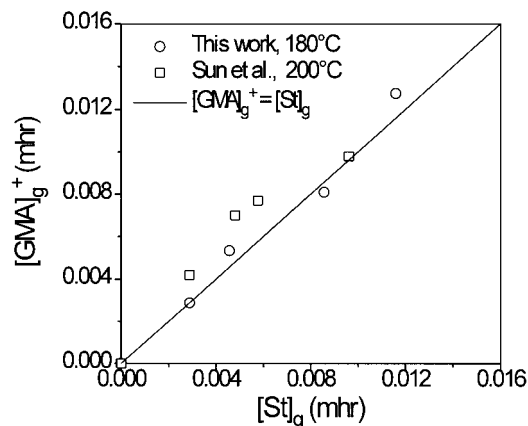


Figure 13. Quantitative analysis of the contribution of adding St to GMA grafting yield increase: experimental data vs. prediction. (—): $[GMA]_g^+ = [GMA]_g^{St} - [GMA]_g^0 = [St]_g$. (○): Data from this work, grafting temperature = 180°C; (□): data from Sun et al.,¹ grafting temperature = 200°C. Note that all the quantities are in “mhr,” which stands for mol per hundred grams of resin (PP).

based on the proposed reaction mechanism and the experimental data in terms of the graft composition, the question remains of why whereas St is more reactive than GMA toward PP macroradicals, the amount of St grafting is lower than that of GMA grafting in a mixture of GMA + St in which the amount of St is equal to or greater than that of GMA. Also, the proposed grafting mechanism is inconsistent with the fact that the reactivity between the GMA radical and St should be similar to that between St radical and GMA (the reactivity ratios of GMA and styrene are 0.53 and 0.44, respectively, at 60°C¹¹; and they are 0.63 and 0.34, respectively, at 65°C¹²; data at higher temperatures are not available), the GMA free radical is expected to react with St as well.

CONCLUSIONS

In this article, we have discussed in detail the “comonomer concept,” which was originally developed by Sun et al.^{1,2} for promoting the otherwise very weak melt free radical grafting reactivity of glycidyl methacrylate (GMA) onto polypropylene (PP). The idea was to add a second monomer (also called a comonomer), which could react with PP macroradicals more rapidly than the monomer (GMA), and the resulting macroradicals would then copolymerize readily with the monomer. In this way, instead of being grafted directly onto

PP, the monomer would be attached to PP via the macroradicals of the comonomer.

This study has provided a larger body of experimental data together with a detailed analysis of how St contributes to the melt free radical grafting of GMA onto PP. Experimental results strongly support the early proposed mechanism that when St is added to the PP/GMA/peroxide grafting system, the dominating mechanism is that St reacts first with the PP macroradicals to form more stable styryl macroradicals. GMA is then grafted onto PP via copolymerization with these styryl macroradicals. This mechanism allows not only enhancing of the GMA grafting yield, reducing PP chain degradation, but also significantly speeds up the grafting rate to within less than 1 or 2 min in a batch mixer. This time includes that necessary for melting the polymer and homogenizing the grafting system. As a matter of fact, the grafting rate is so much accelerated by the addition of St that melting and homogenization become the controlling step. This explains why the free radical grafting of GMA in a corotating twin-screw extruder occurs primarily in the melting zone.^{2,8}

We have shown that the fundamental basis of the comonomer concept is the reactivity ratio of free radical copolymerization. Adding a comonomer to a polymer–monomer grafting systems will be beneficial only if it is capable of reacting more easily with polymer free radicals and the resulting free radicals are able to copolymerize readily with the grafting monomer. St is a good comonomer for the free radical grafting of GMA onto PP because it meets both criteria mentioned above. It is expected that the use of St as comonomer will be beneficial for grafting other monomers onto PP or other types of polymers.¹³ In the future, it will also be important to be able to replace St by other comonomers because the use of St is not recommended for certain applications.

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